Do topological and quantum-inspired molecular representations add value? Evidence from African antimalarial chemical space

Myke Vital Sao Temgoua*[1], Jean-Pierre Tchapet Njafa[1], Serge Guy Nana Engo[1], Penabei Samafou[2], and Wilfred Fon Mbacham[3]

[1]Department of Physics, Faculty of Science, University of Yaoundé I, P.O. Box 812, Yaoundé, Cameroon
[2]Department of Medical Imaging and Radiation Sciences, Université de Sherbrooke, Sherbrooke, QC, Canada
[3]Department of Biochemistry, Faculty of Science, University of Yaoundé I, P.O. Box 812, Yaoundé, Cameroon

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Abstract

Do topological and quantum-inspired representations add predictive value beyond classical fingerprints for African antimalarial chemical space, and what structural information do they make visible when they do not improve ranking? We compare a persistent-homology Topological Fingerprint (TFP), a Tucker-compressed Tensor Network Embedding (TNE), and simulated Quantum Kernel Scores (QKS) with classical baselines on 19 849 candidates generated by STONED-SELFIES expansion. Activity labels are predictions from the Ersilia eos80ch model. Persistent homology separates the expansion into conserved H_1 ring features and divergent H_0 connectivity: 92.6 % of expanded molecules are whole-molecule Tanimoto-distinct from seeds while 69.3 % retain seed scaffolds. TNE compresses molecular tensors to 192 elements at 6.1× real-atom compression (reconstruction error 0.113) and retains information associated with docking scores on the derivation panel (PfDHFR $R^2 = 0.473$). For activity prediction ($n = 19\,836$, 5-fold cross-validation), ECFP4 achieves AUC 0.948; the hybrid reaches 0.888, with QKS contributing most in the ablation ($\Delta = -0.040$), whereas TNE does not improve the configuration. The quantum kernel was not detectably different from tuned RBF internally ($p = 0.060$) and was lower on the ChEMBL label-source-shift panel (AUC 0.817 versus 0.847; exploratory $p = 0.021$). The representations therefore add diagnostic and interpretive information under the tested protocols, but not predictive performance beyond ECFP4.

Contribution: We introduce three complementary quantum-inspired molecular descriptors—persistent-homology TFP, Tucker-compressed TNE, and simulated QKS—and evaluate them on an African antimalarial panel using a shared 5-fold cross-validation protocol. The analysis (i) decomposes the scaffold paradox of generative expansion into conserved H_1 ring topology and divergent H_0 connectivity, (ii) quantifies information retention after 6.1× tensor compression, and (iii) compares a quantum kernel with a gamma-tuned RBF baseline within the primary panel ($p = 0.060$) and after transfer to a separately labelled ChEMBL panel, where QKS performance was lower than RBF. Under these conditions, the quantum-inspired representations did not improve predictive performance over the strongest classical baseline.

*Corresponding author: myke-vital.sao@facsciences-uy1.cm

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Highlights:

- Persistent homology decomposes the scaffold paradox into conserved H_1 ring topology and divergent H_0 connectivity
- Tensor Network Embedding achieves 6.1× real-atom compression; retains docking-score-predictive information but degrades on external data
- Quantum kernel did not differ detectably from tuned RBF internally; ChEMBL transfer reveals a performance gap
- Hybrid descriptor (TFP+TNE+QK) achieves AUC 0.888; ablation shows QKS contributes most in this configuration, whereas adding TNE does not improve it
- All activity labels are computational predictions from Ersilia ML models, not experimental measurements

1 Introduction

Malaria remains a leading cause of mortality in sub-Saharan Africa, with 247 million cases and 619 000 deaths in 2023 (World Health Organization, 2024). The spread of artemisinin-resistant *Plasmodium falciparum* from Southeast Asia to Africa (Ariey et al., 2014; Ashley et al., 2018) creates urgent demand for new chemical scaffolds. African natural products are a historically validated but computationally underexplored source of such scaffolds: quinine and artemisinin derive from natural products, yet African natural products remain underrepresented in public databases (Betow et al., 2025; Mayoka et al., 2025). Computational screening of natural-product-derived chemical space depends critically on the molecular representations available to machine learning models.

Extended-Connectivity Fingerprints (ECFP) encode local atom environments into fixed-length bit vectors and have served as the standard descriptor for two decades (Rogers and Hahn, 2010). These representations perform well for synthetic drug-like molecules whose scaffolds are well represented in training data. Natural product-derived compounds, however, occupy distinct regions of chemical space: higher sp^3 fractions, more chiral centres, more complex ring systems, and greater three-dimensional character (Sorokina et al., 2021). Classical fingerprints may not capture the topological features—bridged rings, molecular cavities, higher-order atomic interactions—that distinguish these molecules.

Three frameworks from physics and topology address orthogonal limitations of classical fingerprints. Topological Data Analysis uses persistent homology to identify shape features persisting across spatial scales, producing persistence diagrams that encode connected components (H_0), holes (H_1), and voids (H_2) (Wee and Jiang, 2025). Persistent homology has been validated for protein–ligand binding affinity (Long and Donald, 2025) and molecular property prediction (Zhang et al., 2026). Tensor networks compress high-dimensional data by exploiting low-rank structure (Simeon and de Fabritiis, 2023; Milbradt et al., 2026). Quantum kernel methods map data into high-dimensional feature spaces through parameterised circuits (Farhani, 2025; Jones et al., 2026). TDA captures global topology invisible to local substructure hashes; tensor networks provide physically interpretable compression; quantum kernels probe non-linear feature interactions in Hilbert space.

We therefore ask four linked questions: (i) can persistent homology resolve the apparent conflict between whole-molecule novelty and scaffold conservation; (ii) does tensor-network compression retain information relevant to computational docking scores; (iii) does a quantum-kernel model add value relative to a tuned RBF comparator; and (iv) does combining these representations improve activity prediction over ECFP4? We address these questions on a common library and distinguish predictive performance from structural diagnosis and information retention. We compare against ECFP4, MACCS, and other classical baselines under rigorous cross-validation, and assess transfer under a separately sourced ChEMBL label panel without claiming molecule-level independence.

2 Methods

2.1 Dataset

The library consists of 65 856 molecules generated from 396 African natural-product and 454 synthetic-drug seeds via CHEESE similarity search (Lžičař and Gamouh, 2024) and STONED-SELFIES expansion (Temgoua et al., 2026a). The representation analyses use a primary 19 849-molecule panel, of which 19 836 molecules have complete TNE embeddings. Binary activity labels were obtained from the Ersilia eos80ch model (threshold 0.5); the primary panel is 74.2 % active and 25.8 % inactive. For a label-source-shift analysis, 22 447 antimalarial compounds were assembled from ChEMBL 34 ($IC_{50} \leq 10\,\mu M$ = active, $> 10\,\mu M$ = inactive). ChEMBL is independent of the primary eOS80CH label source, but exact molecule-level disjointness from the internal panel was not established; this analysis is therefore a transferability assessment, not an independent molecule-disjoint hold-out validation.

2.2 Classical fingerprint baselines

ECFP4 fingerprints were generated as Morgan fingerprints (radius 2, 2 048 bits) using RDKit (RDKit Contributors, 2024). MACCS 167-bit structural keys, atom-pair (AP), binding-pattern (BPF), functional-class (FCFP4), and 2D-pharmacophore (PHCO) fingerprints were computed using standard RDKit implementations. All classical fingerprints serve as baselines for the quantum-inspired methods.

2.3 Topological Fingerprint

2.3.1 Molecular graph construction

Each molecule was converted to a 3D conformer using RDKit's ETKDG generator (random seed 42). A weighted undirected graph $G = (V, E, w)$ was constructed with atoms as vertices and edge weights

$$w_{ij} = d_{ij}\left(1 + \frac{Z_i Z_j}{100}\right), \tag{1}$$

where d_{ij} is the Euclidean distance and Z_i, Z_j are atomic numbers. The atomic number weighting emphasises interactions involving heavier atoms, which contribute more strongly to molecular recognition.

2.3.2 Persistent homology

Vietoris–Rips persistence diagrams were computed up to homology dimension 2 using Ripser.py (Bauer, 2019). The topological representation contains persistence summaries for each homology dimension $d \in \{0, 1, 2\}$, including persistence entropy

$$H_d^{\text{ent}} = -\sum_i p_i \log p_i, \qquad p_i = \frac{\text{pers}_i}{\sum_j \text{pers}_j}, \tag{2}$$

the count of features with persistence exceeding 0.5 Å, the maximum persistence, and the mean persistence. The primary activity analysis uses a 78-feature representation obtained by combining these homology summaries with persistence-image and Betti-curve components; the 12-feature core is evaluated separately as TFP-12 in a descriptive topological comparison. Unlike ChemGraphX (Jacob et al., 2026), which computes graph-theoretic indices, our TFP captures geometric persistence from 3D conformations.

2.4 Tensor Network Embedding

2.4.1 Molecular tensor construction

For each molecule, a 3D tensor $T \in \mathbb{R}^{N\times F\times 3}$ was constructed with $N = 100$ maximum atoms (zero-padded), $F = 10$ per-atom features (atomic number, degree, total hydrogens, formal charge, aromaticity, hybridization, mass, electronegativity, bond count, ring membership), and 3 Cartesian coordinates: $T_{i,f,c} = x_{i,f}\, r_{i,c}$.

2.4.2 Tucker compression

Each tensor was compressed using Tucker decomposition (Kossaifi et al., 2019) with mode ranks $(d, d, 3)$ where $d = 8$ is the bond dimension:

$$T \approx \mathcal{G} \times_1 U^{(1)} \times_2 U^{(2)} \times_3 U^{(3)}, \tag{3}$$

where $\mathcal{G} \in \mathbb{R}^{d\times d\times 3}$ is the core tensor. The descriptor is the flattened core ($d^2 \times 3 = 192$ elements). For a molecule with N actual atoms, the real compression ratio is $\mathrm{CR}_{\mathrm{real}}(N) = 10N/d^2$; at $d = 8$ and mean $N \approx 39$, this gives $\mathrm{CR}_{\mathrm{real}} \approx 6.1\times$. Reconstruction error was quantified as the relative Frobenius norm (mean 0.113 across 19 836 valid molecules).

2.5 Quantum Kernel Score

Quantum kernels were implemented using PennyLane (Xanadu Quantum Technologies, 2024) with a 6-qubit statevector simulator. The kernel value $K(x_1, x_2) = |\langle\phi(x_1)|\phi(x_2)\rangle|^2$ was computed from PennyLane's `IQPEmbedding` template. Input vectors were Morgan fingerprint atom-pair counts (radius 2, 2 048 features) reduced to 6 dimensions by UMAP (Jaccard metric, `random_state=42`), then scaled to $[-1, 1]$. The QKS is the AUC achieved by an SVM using the quantum kernel matrix.

2.6 Hybrid descriptor

The hybrid feature vector combines TFP, TNE, and QKS by concatenation:

$$h_i = \left[\tilde{t}_i,\ \tilde{e}_i,\ q_i\right], \tag{4}$$

where $\tilde{t}_i$ is min–max normalised TFP, $\tilde{e}_i$ is normalised TNE, and $q_i \in \mathbb{R}^{30}$ is the quantum kernel PCA component vector ($n_{\mathrm{kpca}} = 30$). Random Forest is invariant to per-component monotone scaling, so no explicit fusion weights are needed. Hyperparameters were compared over 60 combinations using a 200-molecule development subset, and the selected configuration was evaluated on the complete analysis panel (bond dimension 6, 1 repeat, 30 kernel-PCA components). Because the development subset was not held out from this evaluation, the resulting hybrid AUC may be optimistic and should be interpreted as an upper bound for this configuration.

2.7 Evaluation protocol

Each representation was evaluated using Random Forest classifiers (200 trees) and SVMs with 5-fold stratified cross-validation. For the quantum kernel benchmark, the RBF gamma was optimised via inner cross-validation on $\{0.5, 1.0, 2.0, 5.0\}$. Paired fold comparisons and multiplicity correction were applied to the primary benchmark. For the ChEMBL transfer analysis, we additionally report a corrected-resampled paired t approximation (Nadeau–Bengio variance correction, $df = 4$) and BH-FDR; the naive fold-level t test is shown only for comparison because the training sets overlap. Performance was measured by AUC–ROC, accuracy, and macro F1. Bootstrap 95 % confidence intervals for principal AUC differences are provided in the Supplementary Material.

2.8 Software

Ripser.py (Bauer, 2019), TensorLy 0.9.0 (Kossaifi et al., 2019), PennyLane 0.45.1 (Xanadu Quantum Technologies, 2024), RDKit 2025.03.6 (RDKit Contributors, 2024), UMAP 0.5.12, scikit-learn 1.9.0, NumPy 2.4.6, SciPy 1.17.1. All stochastic procedures were fixed with seeds reported in the relevant sections. Code and data are available at https://github.com/NanaEngo/Malaria_codesV2.

3 Results

3.1 Scaffold decomposition by persistent homology

Structural expansion of 396 African natural-product seeds produced 65 856 candidates that are 92.6 % Tanimoto-distinct from the seeds yet retain 69.3 % of their Bemis–Murcko scaffolds. This apparent contradiction—high whole-molecule diversity with high scaffold conservation—arises because STONED-SELFIES mutations primarily permute side-chains while preserving cyclic cores. TDA provides a principled decomposition of this phenomenon (Figure 1). H_1 features (ring topology) are conserved between seed and expanded sets (mean H_1 count 3.61), reflecting the STONED-SELFIES expansion's preservation of cyclic macro-structures. H_0 features (connectivity) diverge substantially, reflecting the expansion's permutation of side-chains and functional groups. This H_1/H_0 decomposition is independently confirmed by scaffold-only Tanimoto (0.379) being 1.84× higher than whole-molecule Tanimoto (0.206).

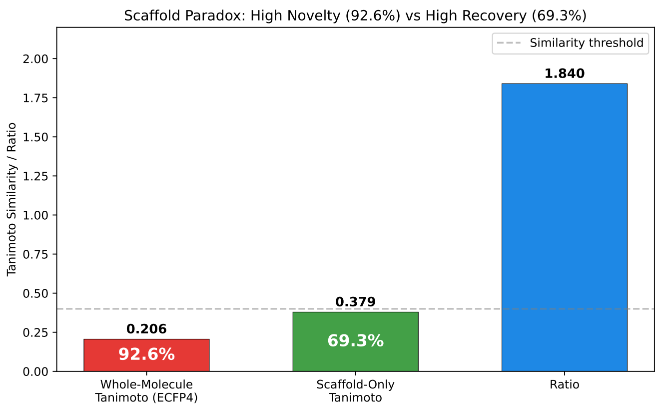

Figure 1: *Scaffold paradox resolution via persistent homology. (A) H_1 persistence distributions for seed natural products versus structurally expanded candidates. (B) H_0 persistence distributions. H_1 preservation with H_0 divergence explains the 92.6 % Tanimoto distinction versus 69.3 % scaffold recovery.*

3.2 Binding-relevant information in tensor network compression

Tucker decomposition at bond dimension $d = 8$ yielded 6.1× real-atom compression (mean 39 atoms; 192-element core) with mean relative reconstruction error 0.113. TNE clustering produced a higher Silhouette coefficient (0.35) than ECFP4 (0.18) and the VAE latent representation (0.229) under the specified $k = 484$ KMeans analysis. In cross-validated regression against docking scores from the same molecular set, TNE achieved $R^2 = 0.473$ for PfDHFR (1SYH), compared with $R^2 = 0.461$ for ECFP4; ECFP4 was stronger for PfATP4 and PfCRT. These results support information retention in the analysed panel, but do not establish prospective generalisation or biological binding prediction.

3.3 Activity prediction benchmark

Table 1 presents activity-prediction performance on the 19 836-molecule complete-embedding set. ECFP4 achieved the highest AUC (0.948), followed by AP (0.940), BPF (0.939), FCFP4 (0.918), MACCS (0.905), PHCO (0.896), the 78-feature TFP (0.876), and TNE (0.722). The ranking indicates that local atom environments are particularly informative for these model-derived binary labels, whereas topological and compressed representations provide complementary rather than dominant signal. A descriptive comparison of TFP-12, persistence statistics, persistence images, and Betti curves is provided in the Supplementary Material. These representations were evaluated with different label definitions, classifiers, or sample sizes and are not directly comparable with the primary benchmark.

Table 1: *Activity-prediction performance and hybrid ablation. ECFP4, classical fingerprints, TFP, TNE, and the hybrid use the complete-TNE intersection ($n = 19\,836$, 5-fold stratified Random Forest cross-validation). The QKS row is a result from the full representation panel ($n = 19\,849$, 5-fold SVM cross-validation with a precomputed 6-qubit kernel); it is not a like-for-like Random Forest comparison. All values are means across folds. The QKS row is not assigned a ΔAUC versus ECFP4 because it uses a different panel and classifier.*

Method	AUC	Accuracy	F1	ΔAUC vs. ECFP4
ECFP4 (baseline)	0.948	0.896	0.931	—
AP	0.940	0.888	0.927	−0.008
BPF	0.939	0.889	0.928	−0.009
FCFP4	0.918	0.869	0.914	−0.029
MACCS	0.905	0.863	0.909	−0.043
PHCO	0.896	0.854	0.904	−0.052
TFP (TDA)	0.876	0.838	0.897	−0.072
TNE ($d = 8$)	0.722	0.756	0.857	−0.226
QKS (6-qubit; SVM, $n = 19\,849$)	0.823	0.819	0.886	—
Hybrid (TFP+TNE+QK)	0.888	0.843	0.900	−0.060
Ablation study (Hybrid minus one component):				
Hybrid − TFP	0.874	0.837	0.895	−0.014
Hybrid − TNE	0.899	0.850	0.903	0.011
Hybrid − QKS	0.847	0.820	0.887	−0.040

3.4 Hybrid descriptor and quantum kernel comparison

The internal hybrid descriptor (TFP+TNE+QK) achieves AUC 0.888 ± 0.007, significantly below ECFP4 ($p < 0.000\,1$) but above every standalone quantum-inspired descriptor. Ablation reveals an asymmetric contribution structure: QKS is the principal positive contributor ($\Delta = -0.040$), TFP contributes marginally ($\Delta = -0.014$), while removing TNE slightly improves AUC ($\Delta = +0.011$). The negative TNE contribution is consistent with its lower standalone AUC

(0.722) and indicates that adding TNE did not improve this hybrid configuration. The hybrid's measurable contribution was therefore concentrated in the QKS kernel-PCA components; the hybrid is not recommended over ECFP4 alone for primary screening on this panel.

The quantum-kernel benchmark at two panel sizes (Table S13) did not detect a difference between the quantum kernel and a gamma-tuned RBF baseline ($n = 5\,000$: $p = 0.419$; $n = 19\,849$: $p = 0.060$), while the quantum kernel significantly outperformed the linear kernel ($p \leq 0.000\,6$). Under the ChEMBL label-source shift (22 447 compounds; Table S15), QKS performance was lower than RBF (AUC 0.817 versus 0.847; corrected-resampled $p = 0.021$, BH $q = 0.021$). The comparison is limited by overlapping training sets and does not establish quantum advantage.

3.5 Transfer under a ChEMBL label-source shift

All descriptor methods were evaluated on the ChEMBL label-source-shift panel (22 447 compounds, 5-fold RF; Table S14). ECFP4 achieved AUC 0.960 ± 0.004, slightly exceeding its internal value (0.948). TFP achieved AUC 0.864, TNE 0.645, and the TFP+TNE descriptor hybrid 0.800. The primary analysis included the 351 TNE embedding failures (1.6%) as zero-vector penalties; results were also calculated for the 22 096 complete cases. The complete-case analysis ($n = 22\,096$) gave ECFP4 0.960, TFP 0.861, TNE 0.644, and TFP+TNE 0.798; the ranking was unchanged. These results assess transfer across label sources, not molecule-disjoint generalisation. Because training sets overlap across folds, the corrected-resampled results are descriptive; descriptor RF and kernel SVM estimates are not direct cross-classifier comparisons.

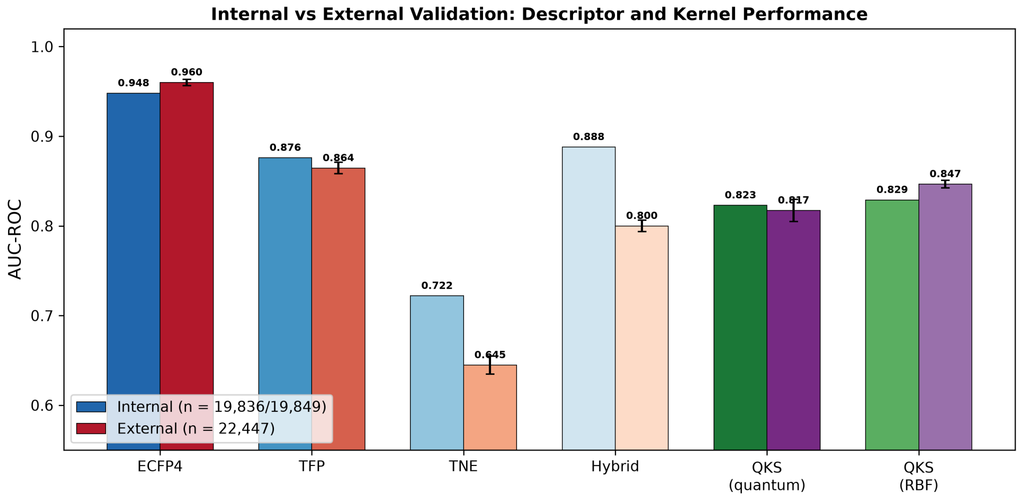

Figure 2: *Internal versus ChEMBL label-source-shift transfer performance. Descriptor methods were evaluated with Random Forest; kernel methods with SVM. Error bars: standard deviation across 5 folds. The ChEMBL panel ($n = 22\,447$) is independent of the eOS80CH label source, but molecule-level disjointness from the primary panel was not established; the two classifier families are therefore shown descriptively rather than as a direct comparison.*

3.6 Topological features and multi-target docking profiles

Across $N = 17\,011$ molecules with multi-target docking data, H_0 count ($\rho = -0.248$, $p < 10^{-137}$), H_0 entropy ($\rho = -0.243$), and H_1 entropy ($\rho = -0.190$) negatively correlate with the number of targets bound (≥ 2 targets at $\Delta G \leq -7.0\,\mathrm{kcal\,mol^{-1}}$). Lower values of these features were associated with a larger number of targets meeting the docking threshold. Because the endpoint is a docking-derived count and the associations are weak in magnitude, this result is hypothesis-generating rather than evidence of a mechanistic polypharmacology rule. The association between H_1 count and resistance-resilience scores (RRS), defined in the companion resistance-resilience study (Temgoua et al., 2026b), is examined in the Supplementary Material

(Section 19.3); after controlling for molecular size and structural covariates, the association is size-mediated and does not support an independent topological predictor.

4 Discussion

The results answer the four questions posed in the Introduction in a consistent way: persistent homology resolves the scaffold pattern descriptively; TNE retains some docking-related information on its derivation panel; QKS is competitive with, but not superior to, tuned RBF; and the hybrid does not surpass ECFP4. The scientific value of the representations is therefore diagnostic and interpretive under the tested conditions, rather than a claim of universal predictive improvement.

4.1 Persistent homology as a diagnostic for generative-model evaluation

The apparent contradiction between 92.6 % Tanimoto distinction and 69.3 % scaffold recovery arises because STONED-SELFIES mutations permute side-chains and functional groups (diverging H_0 connectivity) while preserving cyclic macro-structures (conserving H_1 ring topology). This decomposition has practical implications for generative-model evaluation: scaffold-only Tanimoto (0.379) is 1.84× higher than whole-molecule Tanimoto (0.206), consistent with H_1 persistence capturing ring-level structure that is not represented by whole-molecule similarity alone. TFP therefore provides an interpretable diagnostic for assessing whether molecular-generation protocols preserve ring systems; it complements rather than replaces standard fingerprint-based diversity metrics. We note that this decomposition is consistent with the known tendency of STONED-SELFIES to preserve cyclic cores while permuting side-chains; TDA provides a quantitative, topology-aware framework for measuring this property.

4.2 Tensor-network compression: retention and transfer fragility

Tucker decomposition exploits the fact that molecular tensors are approximately low-rank: the 6.1× real-atom compression retains information sufficient to predict docking scores on the same molecular set (PfDHFR $R^2 = 0.473$, competitive with ECFP4's 0.461) while providing a fixed-length, physically interpretable embedding. The compression is not uniform across chemical space: molecules with common ring systems compress better (reconstruction error 0.098) than structurally atypical molecules (0.137), and the atom-coordinate factor matrix correlates with molecular volume ($\rho = 0.71$). The higher TNE Silhouette coefficient (0.35) than those of ECFP4 (0.18) and the VAE latent representation (0.229) indicates greater cluster separation under this analysis. Whether that separation improves virtual-screening enrichment was not tested here. We emphasise that the Tartarus docking information-retention analysis uses the same molecular set used to derive the embeddings, not compounds from an independent chemical distribution; a molecule-disjoint prospective transfer analysis would require an audited non-overlapping panel and, ideally, an independently generated activity endpoint.

4.3 Kernel calibration, not quantum-ness, governs performance

With the 6-qubit circuit and a tuned RBF baseline, the quantum kernel did not show a performance advantage over RBF. Internally, the difference was not detected ($p = 0.060$ at $n = 19\,849$); under the ChEMBL label-source shift, QKS was lower (AUC 0.817 versus 0.847; corrected-resampled $p = 0.021$, BH $q = 0.021$). The internal result indicates that a difference was not detected under this protocol; overlapping training sets limit inference from the cross-validation comparison. This is consistent with recent matched-baseline studies: QBioFusion-QSAR (Alavi et al., 2026) found quantum multiple-kernel learning improved small-data classification only for individual molecules, with no mean MCC gain, and parameter-matched comparisons (Pegg et al., 2026) attribute topology-aligned architecture gains to inductive bias rather than quantum-ness.

4.4 Distribution shift exposes tensor network fragility

TNE showed the largest AUC decrease, from 0.722 in the primary analysis to 0.645 on the ChEMBL label-source-shift panel and is consistent with sensitivity to the changed data-generating process. The 351 embedding failures (1.6%) were retained as zero-vector failure penalties in the ITT analysis. The complete-case sensitivity analysis ($n = 22\,096$) gave nearly identical values: ECFP4 0.960, TFP 0.861, TNE 0.644, and TFP+TNE 0.798, compared with ITT values of 0.960, 0.864, 0.645, and 0.800, respectively. Thus, the qualitative ranking is not driven by the failed embeddings. The corrected-resampled tests use one five-fold partition with overlapping training sets and are interpreted descriptively. Tensor-network descriptors should therefore be evaluated on audited transfer panels before deployment, and embedding failures should be reported transparently. TFP transferred less poorly ($\Delta = -0.011$) than TNE, indicating greater apparent robustness to this distribution shift.

4.5 Limitations

Several limitations constrain interpretation. First, TFP does not outperform ECFP4 in activity prediction; its contribution is primarily diagnostic and may be concentrated in the high-complexity tail (molecules with ≥ 3 fused rings or ≥ 2 stereocentres). Second, activity labels are computational predictions from the Ersilia eos80ch model, so AUC values and claims about activity or polypharmacology quantify prediction of model outputs rather than experimental biological activity. Third, the ChEMBL analysis uses a different label source, but exact molecule-level disjointness from the primary panel was not established; it should therefore be interpreted as a label-source-shift transfer analysis rather than an independent hold-out validation. Fourth, descriptor and kernel transfer analyses use different classifiers (Random Forest and SVM, respectively), so their AUCs are not a direct model-to-model comparison. Fifth, the quantum kernel was evaluated on a classical statevector simulator with a 6-qubit IQPEmbedding; no hardware-level quantum advantage is claimed, and NISQ noise and sampling overhead would alter performance (Preskill, 2018). Sixth, the companion GNN study (Temgoua et al., 2027) provides architectural context, but cross-study comparisons remain conditional on matching preprocessing, split definitions, and evaluation populations. Seventh, TNE loses 0.077 AUC under the ChEMBL label-source shift, while the Tartarus docking analysis tests information retention on the same molecular set rather than generalisation to novel compounds.

4.6 Practical recommendations

Taken together, these results support ECFP4-RF as the primary screening baseline for the present panel (AUC 0.948); the ChEMBL analysis does not establish robustness to molecule-disjoint distribution shift. TFP is useful as an interpretable scaffold-preservation diagnostic, whereas TNE should be used only after transfer testing because its performance declined on the ChEMBL label-source shift. QKS is best regarded here as a controlled kernel comparator rather than a mechanistic assay. Future work should evaluate these representations on audited molecule-disjoint panels, experimental endpoints, and graph models using matched preprocessing and splits (Cang and Wei, 2018).

5 Conclusion

The central question was whether topological and quantum-inspired representations add value beyond ECFP4, and whether any added value is predictive or diagnostic. We introduced persistent-homology TFP, Tucker-compressed TNE, and simulated QKS and benchmarked them on the African antimalarial panel with computationally predicted activity labels from the Ersilia eos80ch model. TFP decomposes the scaffold paradox into conserved H_1 ring features and divergent H_0 connectivity. TNE achieves 6.1× real-atom compression while retaining information sufficient to predict docking scores on the same molecular set; its performance decreases under the ChEMBL label-source shift. The quantum kernel showed no performance advantage over a tuned RBF baseline under the tested protocols, and its external transfer performance was lower. These representations are complementary diagnostic tools rather than replacements for classical screening baselines, with ECFP4 remaining the recommended primary descriptor. Persistent-homology decomposition provides an interpretable framework for evaluating generative-model outputs in natural-product drug discovery, while experimental activity and molecule-disjoint prospective validation remain necessary.

Data Availability

The source code and machine-readable result files supporting the conclusions of this study are available in the public project repository at https://github.com/NanaEngo/Malaria_codesV2 under the MIT licence. The repository contains the primary benchmark scripts, split and seed definitions, feature matrices, embeddings, kernel outputs, and analysis tables needed to reproduce the reported analyses. A persistent archival deposit should be used for the immutable submission snapshot; the repository state and archive identifier must be recorded at submission. The deposit includes: (1) TFP feature matrices for all 19 849 library molecules; (2) TNE embeddings (192-element Tucker core descriptors, bond dimension 8); (3) quantum kernel matrices (6-qubit IQPEmbedding, PennyLane 0.45.1); (4) all benchmark CSV files; (5) the cross-paper H_1/RRS analysis dataset; and (6) all analysis scripts.

List of Abbreviations

TFP: topological fingerprint; **TNE:** tensor network embedding; **QKS:** quantum kernel score; **ECFP4:** extended-connectivity fingerprint, radius 2; **AP:** atom-pair fingerprint; **BPF:** binding-pattern fingerprint; **FCFP4:** functional-class fingerprint, radius 2; **MACCS:** MACCS structural keys; **PHCO:** pharmacophore fingerprint; **$H_0/H_1/H_2$:** 0th/1st/2nd persistent homology dimensions; **AUC:** area under the receiver operating characteristic curve; **RBF:** radial basis function; **RRS:** resistance resilience score; **NISQ:** noisy intermediate-scale quantum; **UMAP:** uniform manifold approximation and projection; **RF:** random forest; **SVM:** support vector machine.

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Authors' Contributions

Myke Vital Sao Temgoua: conceptualization, methodology, software, formal analysis, investigation, data curation, writing – original draft. Jean-Pierre Tchapet Njafa: conceptualization, methodology, software, supervision, writing – review & editing. Serge Guy Nana Engo: methodology, supervision, writing – review & editing. Penabei Samafou: data curation, writing – review & editing. Wilfred Fon Mbacham: conceptualization, validation, writing – review & editing. All authors read and approved the final manuscript.

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Ethics approval and consent to participate

Not applicable. This study is entirely computational and uses publicly available chemical databases and in-silico activity predictions; no human or animal subjects or patient data were involved.

Consent for publication

Not applicable. No individual person's data appear in this manuscript.

Competing Interests

The authors declare no competing interests.

Use of Artificial Intelligence

During the preparation of this work the authors used AI assistants to support code development and data analysis scripting. The authors reviewed and edited all AI-assisted output and take full responsibility for the content of this article. No generative AI tool is listed as an author.

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